Do Rich Guys Finish First ?
Analyzing correlation between wealth and electoral performance in the Indian general elections 2004-2014

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Abstract

Using a database of assets declarations of candidates contesting in the three most recent Indian general elections conducted over the last decade, we show that the distribution of their wealth follows a universal scaling form independent of the year, as well as the states and the parties to which the candidates belong. We also observe that the set of winners, as well as, that of the "serious candidates" (contenders) have asset distributions which deviate significantly from those of the remaining candidates, a worrying aspect in context of the apparently representative nature of democracies.

Keyword: Inequality, Elections, Wealth, Universality, Lognormal distribution

Inequality is very commonly observed in socio-economic phenomena, appearing as highly skewed distributions of attributes such as wealth and popularity, e.g., in terms of electoral success. In democratic societies that are ostensibly based on the principle of equality of all citizens, the relation between wealth and electoral success has been a contentious subject. One can ask whether in an election where, in principle, any individual regardless of their financial standing can compete, the wealth of the candidate can be a significant factor affecting their electoral performance. While media reports often suggest that money can indeed influence elections, several studies have shown that campaign spending have little apparent impact on electoral outcomes [1, 2]. However, the role of wealth may be much more subtle than that captured simply by considering the amount spent directly in the campaign.

In this work, using the data available publicly about the assets owned by candidates appearing in elections for the Lok Sabha (the lower house of the Indian Parliament) as well as information about their electoral performance, we show that electoral winners are not representative of the population - of all candidates, in lieu of the entire society - either in terms of wealth or wealth inequality. We have also used an entropy-based measure to distinguish the contenders ("serious candidates") from the rest, and find even this segment of candidates have assets distribution that deviate significantly from that of the others. This has been confirmed by

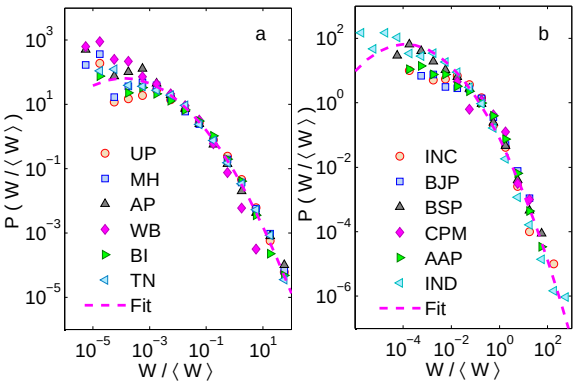

Figure1: Rescaled probability distributions of total assets for candidates contesting in the Lok Sabha election of 2014 for (a) the six largest states in India in terms of the number of elected representatives in the Lok Sabha, viz., Uttar Pradesh (UP), Maharashtra (MH), Andhra Pradesh (AP), West Bengal (WB), Bihar (BI) and Tamil Nadu (TN), and (b) several political parties, including the four largest national parties (INC, BJP, BSP, CPM) and a newly emerged party (AAP), as well as Independents (IND) which include all unaffiliated candidates. Scaling is done by dividing assets by the average for each state in (a) and for each party in (b). The data is described well by a lognormal distribution with $\sigma^2 = 6$ shown by the broken curve.

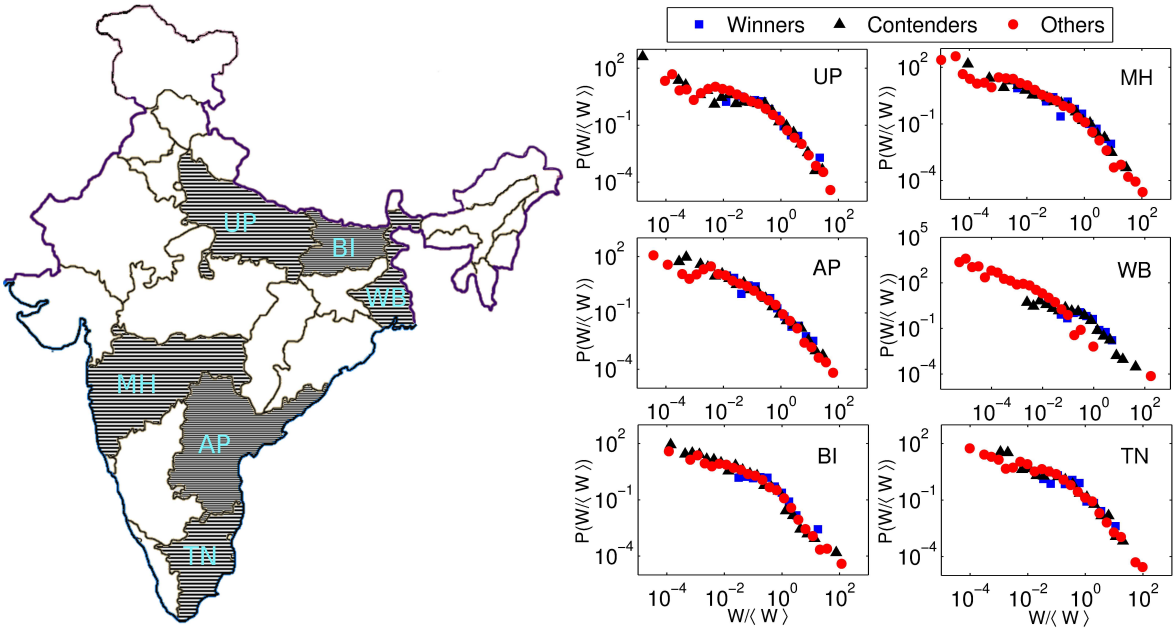

Figure2: Rescaled probability distributions of assets for winners, contenders (*sans* winners) and the remaining candidates contesting in the Lok Sabha election of 2014 for the six largest states in India in terms of the number of elected representatives in the Lok Sabha (indicated in the map).

two-sample Kolmogorov-Smirnov tests carried out on the data for the three sets of candidates, viz., winners, contenders and the rest. This is surprising, especially given that the wealth distribution of all candidates is described very well by a log-normal distribution, which has an universal form once the data is scaled by its mean. Fig. 1 makes the invariant nature of the scaling function apparent as we compare assets distribution of candidates (a) from different states in the country and (b) belonging to different parties. Although these states differ significantly from each other in demographic indicators and the parties differ in their geographical presence, appeal to different classes, etc., once we account for the different mean values, the distribution fits the same quantitative form.

Thus, it appears that, the winners and even the contenders, not only differ from the rest of the candidates in terms of their wealth but are in fact much more homogeneous (in terms of lower wealth inequality) within themselves as compared to the others. In fact, even the winners and contenders significantly differ from each other in terms of their assets distribution. Our work suggests that wealth, while it may not have a simple predictable relation to winning, is nevertheless a robust indicator of one or more decisive factors underlying electoral performance (e.g., whether the person belongs to the sociopolitical elite, caste factors, access to resources, existing client-patron relations etc.). These results raises the question whether elections in a representative democracy do indeed provide a level playing field to all its citizens.

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